FUSION ANYTHING: A GENERALIZED MULTIMODAL FOUNDATION MODEL

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ABSTRACT

Making prediction with multimodal data is widely used in diverse scenarios. Existing multimodal fusion models, once deployed, can only handle predefined modalities (e.g., vision, text and audio) and single task, making it difficult to quickly adapt to new downstream applications. Therefore, a natural yet aggressive question arises - whether there exists a general multimodal fusion model that can be applied to arbitrary modality combinations and arbitrary prediction tasks. We argue that a unified multimodal fusion model should not depend on specific modalities and should instead encode transferable patterns of multimodal correlation. To this end, we propose a simple and effective learning paradigm based on training on large-scale synthetic multimodal datasets generated with Structural Multimodal Causal Models (SMCMs), which formally characterizes the generative processes of real-world multimodal data. Building on this framework, we propose the **Fusion Anything Model (FAM)**, a foundation model for generalized multimodal data fusion. By constructing large-scale synthetic multimodal data with diverse correlation patterns, our model encodes transferable multimodal correlations during training and activates appropriate associations through in-context examples during inference. Extensive experiments over 20 real-world datasets spanning 12 modalities and 11 prediction tasks demonstrate that our model achieves competitive performance with specialized models without task-specific adaptation.

1 Introduction

Making prediction with multimodal data focuses on exploiting synergy across modalities, enabling models to capture richer information and make better predictions Baltrušaitis et al. [2018], Xu et al. [2023], Zhang et al. [2026]. By modeling interactions among heterogeneous modalities, multimodal learning has become an important paradigm for understanding complex real-world data Liang et al. [2023], Zou et al. [2025]. However, existing multimodal fusion models are predominantly developed under a task-specific paradigm, where they are designed for predefined modalities and one specific prediction objective Haque et al. [2020], Zhang et al. [2023], Wei et al. [2020]. Consequently, these models mainly learn fusion strategies tailored to specific scenarios, making it difficult to capture generalized multimodal correlation patterns beyond predefined modality combinations and task settings.

Recent success of foundation models demonstrates that large-scale pretraining over diverse data distributions enables models to learn transferable patterns and acquire generalizable capabilities across diverse tasks and domains Bommasani et al. [2021], Kirillov et al. [2023], Yang et al. [2024], Hollmann et al. [2025], Wang et al. [2026a]. Following this paradigm, multimodal foundation models have emerged to integrate heterogeneous information sources and advance multimodal understanding and prediction Liang et al. [2024], Wang et al. [2026b], Dong et al. [2026]. However, existing models remain largely confined to a limited modality space, focusing on several well-established modalities such as vision, language, and audio while leaving many other heterogeneous modalities largely unexplored. Such modality-specific designs hinder their ability to capture the complex correlations of generalized modalities that naturally arise in real-world scenarios. Accordingly, a fundamental and aggressive question arises: **can we develop a multimodal foundation model that can rapidly adapt to arbitrary modality combinations and prediction tasks (see Figure 1)**?

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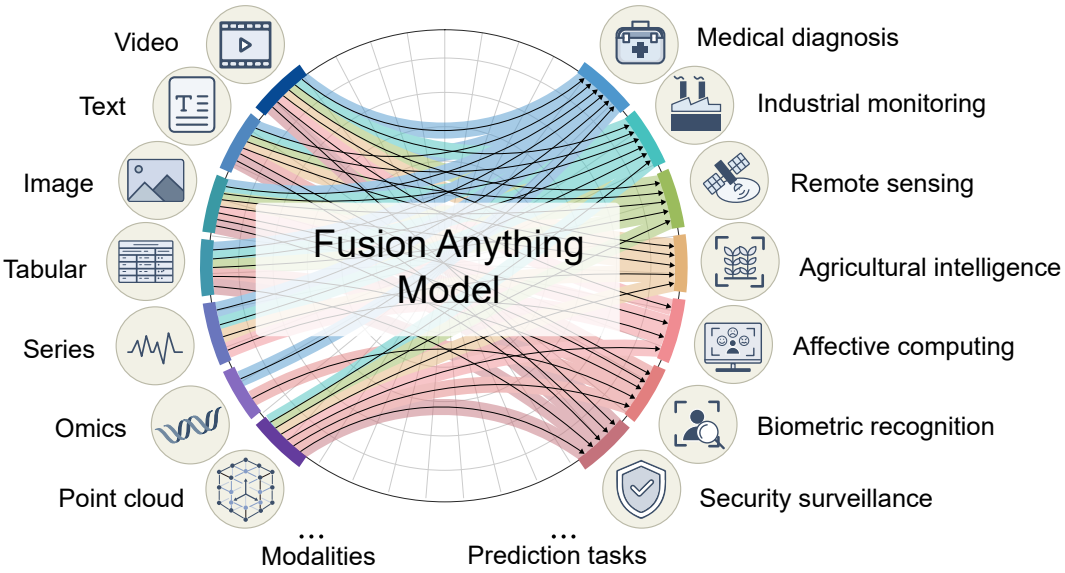


Figure 1: Fusion Anything Model for generalized multimodal prediction. A unified foundation model accommodates arbitrary modality combinations and diverse prediction tasks, enabling direct prediction without task-specific parameter adaptation.

The first challenge of achieving such a generalized multimodal foundation model is the requirement of a large-scale and diverse collection of multimodal data that sufficiently cover the characteristics of real-world data. However, constructing such a collection of real data is inherently challenging. Existing multimodal datasets are highly concentrated on limited modality combinations Xue et al. [2026], Liang et al. [2021], while many other modalities remain scarce due to domain-specific acquisition processes and limited data permission. Moreover, collecting, annotating, and aligning large-scale datasets across diverse domains introduces substantial cost and practical difficulties. These limitations make it difficult to obtain comprehensive training data that cover diverse modality combinations and multimodal correlation patterns. Another challenge lies in the model design, namely how to develop a unified architecture that can adapt to future unseen prediction tasks. Existing multimodal models are predominantly optimized for predefined modalities and task objectives Wang et al. [2026b], Han et al. [2022a], resulting in architectures that are tightly coupled with specific scenarios. Such designs limit their ability to transfer learned multimodal correlation patterns to new tasks and modality combinations.

To address these challenges, in this paper, we propose a new learning paradigm for generalized multimodal fusion based on large-scale synthetic multimodal data. Instead of relying on real multimodal data with limited modality coverage, we construct synthetic multimodal data with Structural Multimodal Causal Models (SMCMs), where complex multimodal correlations naturally emerge through the underlying data generation processes. By training over a broad collection of synthetic multimodal data, our model learns transferable multimodal correlation patterns beyond specific modality combinations and datasets. Building upon this paradigm, we develop Fusion Anything Model (FAM), a multimodal foundation model with the in-context learning capability Hollmann et al. [2025] for diverse multimodal prediction tasks. Through a unified token generation mechanism, our model maps heterogeneous modalities into a unified token space and enables multimodal in-context learning over diverse correlation patterns. During inference, it leverages contextual examples from real datasets to activate task-relevant multimodal correlations and perform prediction without additional parameter updates. Extensive experiments across diverse real-world datasets demonstrate that our Fusion Anything Model outperforms the specialized multimodal fusion models while substantially improving generalization beyond predefined scenarios. Our contributions are summarized as follows:

- We introduce a new learning paradigm for generalized multimodal fusion by constructing large-scale synthetic datasets with diverse SMCMs, enabling general multimodal correlation patterns are pretrained in the fundation model.
- We propose the Fusion Anything Model, a unified foundation model that adapts to diverse modality combinations and prediction tasks through multimodal in-context learning.
- Extensive experiments over 20 real-world datasets demonstrate that the proposed model achieves promising performance compared with specialized models without task-specific adaptation.

2 Related Work

Multimodal Fusion. Multimodal fusion aims to integrate heterogeneous information from multiple modalities for more comprehensive understanding Zhang et al. [2019]. Early research mainly explores feature-level and decision-level fusion strategies. Feature-level fusion combines modality representations before prediction Ramachandram and Taylor [2017], Atrey et al. [2010], while decision-level fusion aggregates predictions from individual modalities, providing modularity but limited cross-modal interaction Han et al. [2022b], Zhang et al. [2023]. With the development of deep learning, intermediate fusion methods introduce separate encoders for each modality and interaction modules to jointly learn modality representations and cross-modal correlations Xu et al. [2023]. Recently, Transformer-based architectures have become the dominant paradigm due to their ability to model complex dependencies through attention mechanisms Liang et al. [2021], Girdhar et al. [2022], Wang et al. [2022]. Existing approaches typically follow either single-stream architectures, which project multiple modalities into a shared representation space, or multi-stream architectures, which retain separate encoders for each modality and perform cross-attention fusion Nagrani et al. [2021]. Furthermore, cross-attention mechanisms have been widely adopted to explicitly capture interactions between heterogeneous modalities and dynamically integrate complementary information Sun et al. [2021], Zhan et al. [2021]. Despite their effectiveness on specific tasks, these methods usually assume predefined modality combinations and require task-specific architectures, limiting their generalization to diverse and unseen multimodal scenarios.

Multimodal Foundation Models. Recent advances in foundation models have shifted multimodal learning toward large-scale pretraining and transferable representations Huang et al. [2026], Dong et al. [2026]. Early multimodal foundation models mainly focus on learning aligned representations between vision and language, with representative works such as CLIP Radford et al. [2021] and ALIGN Jia et al. [2021]. Building upon these advances, multimodal large language models (MLLMs) further integrate visual encoders with large language models to enable multimodal understanding and reasoning, including Flamingo Alayrac et al. [2022], BLIP-2 Li et al. [2023], and LLaVA Liu et al. [2023]. More recent studies have explored unified multimodal architectures by mapping different modalities into shared token spaces and leveraging autoregressive modeling, such as Chameleon Team [2024], Emu3 Wang et al. [2026b], and Janus Wu et al. [2025]. However, existing works are still primarily designed for limited modality combinations and often rely on extensive modality-specific pretraining, making it challenging to achieve generalization across arbitrary modalities and task settings.

Tabular Foundation Models. TabPFN Hollmann et al. [2025] is a tabular foundation model pre-trained on large-scale synthetic datasets with diverse causal structures, enabling it to capture general variable dependencies and perform prediction on unseen structured datasets through contextual examples. Subsequent work has further advanced this line of research Lee [2026], Ma et al. [2026]. For example, LimiX Wang et al. [2026a] extends structured-data modeling to multiple tasks within a unified model, including classification, regression, imputation, and data generation. MMPFN Kim et al. [2026] further incorporates image and text representations into TabPFN through modality projector, but requires fine-tuning on individual real-world datasets. Overall, current tabular foundation models are largely centered on structured-data prediction, while extending such transferable modeling capabilities to heterogeneous multimodal data remains underexplored.

3 Method

3.1 Preliminaries

Let $\mathcal{D} = \{(\mathbf{X}_i, y_i)\}_{i=1}^{N}$ be a dataset of N samples with M modalities, where $\mathbf{X}_i = (\mathbf{x}_i^1, \ldots, \mathbf{x}_i^M)$, $\mathbf{x}_i^m \in \mathbb{R}^{d_m}$ is the encoded representation of modality m, and y_i is the target label. Existing fusion methods typically fit a model p_{θ^*} to each dataset, learning correlations specific to its modality combination and prediction task. In this work, we aim to learn a generalized foundation model p_θ which captures transferable multimodal correlation patterns and is applicable to novel multimodal prediction tasks.

3.2 Generation of Synthetic Multimodal Data

Real-world multimodal datasets are limited and costly to collect. Therefore, we construct synthetic datasets with controllable modality combinations and correlation patterns to support the model training across diverse multimodal settings. Figure 2 illustrates the motivation and overview of our synthetic multimodal data generation process. As shown in Figure 2(a), real-world data are commonly collected by observations of a single target by different sensors. Inspired by this observation, Figure 2(b) depicts the basic principle adopted in our synthetic construction: a common root node R is connected to multiple modality nodes M_m through distinct generation mechanisms f_m. At the sample level, these mechanisms transform the underlying latent information into heterogeneous modality representations, naturally inducing cross-modal correlations through their common source.

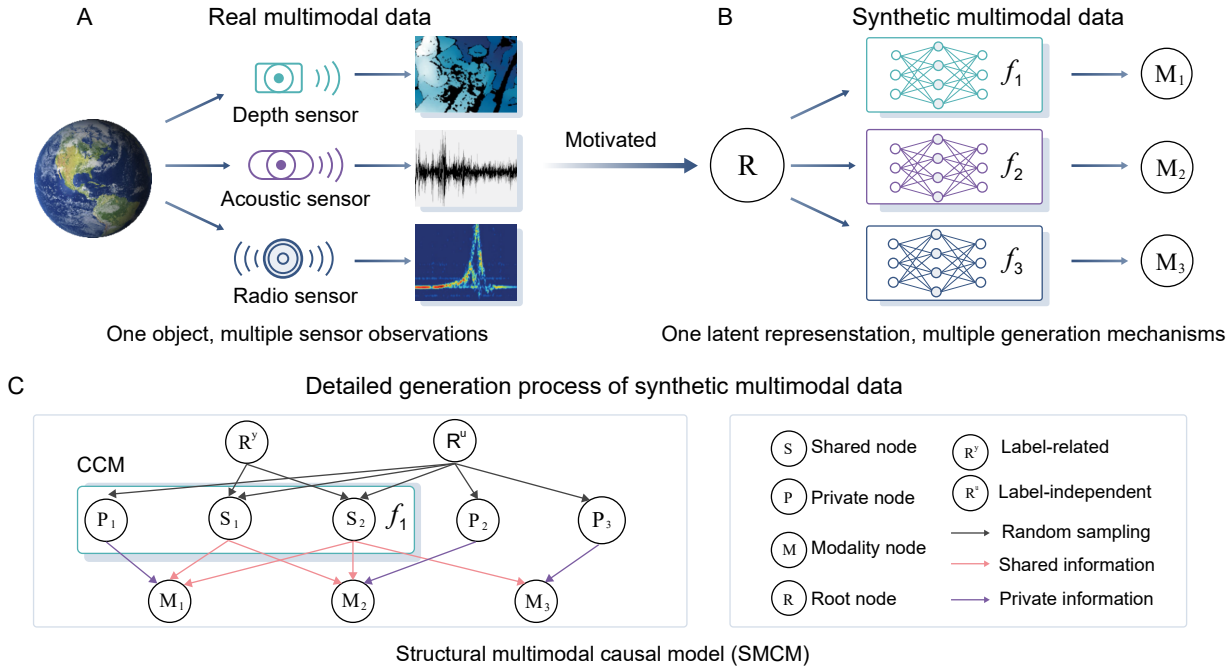


Figure 2: Synthetic multimodal dataset generation. (a) Real-world multimodal observations are produced by different sensors perceiving the same underlying object. (b) Motivated by this process, we generate synthetic modalities from a common latent representation through distinct generation mechanisms. (c) An example SMCM with three modalities M_1, M_2, and M_3. The root representation comprises label-related and label-independent components, and the CCM controls the allocation of latent information through shared and private nodes. Shared nodes provide common information to multiple modalities, while each private node supplies modality-specific variation to its corresponding modality. Varying these node configurations and connections induces diverse multimodal correlation patterns.

However, directly generating each modality from the root provides little control over how information is shared across modalities or retained within individual modalities, making it difficult to cover diverse multimodal correlation patterns with limited generated data. To address this limitation, we propose Structural Multimodal Causal Models (SMCMs) illustrated in Figure 2(c). An SMCM characterizes the multimodal generation process through a structured causal graph comprising a root node, a Correlation Control Module (CCM), and multiple modality nodes. The CCM regulates how latent information contributes to different modalities through shared nodes S_j and private nodes P_m, shared nodes convey selected information to multiple modality nodes, while each private node contributes modality-specific variation only to its associated modality. Varying the configurations of these nodes and their connections to modality nodes enables the generation of synthetic datasets with diverse correlation patterns.

Generation Mechanisms in SMCM. The SMCM organizes multimodal data generation through four types of nodes: a root node R, shared nodes S_j, private nodes P_m, and modality nodes M_m. To define the internal generation mechanisms of the root and modality nodes, we use Structural Causal Models (SCMs) Pearl [2009]. An SCM consists of a directed acyclic graph and a set of structural mechanisms that specify how its variables are generated Hollmann et al. [2025], Wang et al. [2026a]. Within each SCM, each root variable is sampled from either a Gaussian distribution $\mathcal{N}(0, \tau^2_{\text{init}})$ or a uniform distribution $\mathcal{U}(-c_{\text{init}}, c_{\text{init}})$, with the distribution type randomly selected and the scale parameters treated as hyperparameters. Each non-root variable v_a is then generated from its parent variables as follows:

$$v_a = f_a\left(\{g_{a,b}(v_b)\}_{v_b \in \text{Pa}(v_a)}, \epsilon_a\right), \tag{1}$$

where $\text{Pa}(v_a)$ denotes the set of parent variables of v_a, $g_{a,b}(\cdot)$ denotes the edge function that transforms parent variable v_b before its contribution to v_a, $f_a(\cdot)$ denotes an aggregation function that combines the transformed parent variables, and ϵ_a represents an observational noise term. The diversity of variable dependencies is further controlled by the functional forms of the edge and aggregation functions. The edge functions can be instantiated using neural networks with different activation functions (e.g., sigmoid, ReLU, and Tanh), convolutional mappings, or decision-tree functions, while the aggregation function can take the form of mean, weighted, or neural aggregation.

Root Node. The root node R represents an underlying (complete) representation inducing all modalities Zhang et al. [2018], from which the shared, private, and modality nodes are subsequently constructed. We decompose the root representation into label-related and label-independent components to explicitly control task-related information while preserving other latent variations Paige et al. [2017]. Specifically, for sample i, the root representation is defined as:

$$\mathbf{r}_i = (\mathbf{h}_i + \alpha \boldsymbol{\mu}_{y_i})\mathbf{A} = [\mathbf{r}_i^y \| \mathbf{r}_i^u], \tag{2}$$

where $\mathbf{r}_i$ denotes the realization of root node R, $\mathbf{h}_i$ is the initial latent representation of sample i, $\boldsymbol{\mu}_{y_i}$ denotes the label-specific shift associated with class y_i, and α controls the strength of the label-related variation. The transformation matrix $\mathbf{A}$ aligns the label-related variation with a dedicated subspace, yielding the label-related component $\mathbf{r}_i^y$, while the remaining dimensions form the label-independent component $\mathbf{r}_i^u$. These two components are concatenated along the representation dimension, denoted by $\|$, to form the complete root representation $\mathbf{r}_i$.

Shared and Private Nodes. The Correlation Control Module (CCM) consists of shared and private nodes and enables explicit control over the multimodal correlation structures formed during modality generation. As illustrated in Figure 2(c), given the root representation $\mathbf{r}_i = [\mathbf{r}_i^y \| \mathbf{r}_i^u]$, each shared node S_j selects dimensions indexed by Ω_j^y and Ω_j^u from the label-related and label-independent components, respectively. The selected dimensions are then processed through a nonlinear transformation to produce the shared representation $\mathbf{s}_i^j$. Each shared node is connected to multiple modality nodes. In contrast, private node P_m selects dimensions indexed by Ω_m^p from the label-independent component $\mathbf{r}_i^u$ that are not assigned to shared nodes. The selected dimensions are likewise processed by an independent nonlinear transformation to produce the private representation $\mathbf{p}_i^m$ associated exclusively with modality M_m.

Modality Node. Each modality node M_m characterizes the specific modality representation in each synthetic dataset, which integrates information derived from the root node, subsequently the associated shared nodes and private node. For sample i, the representations of all shared nodes connected to M_m are concatenated and denoted by $\mathbf{s}_i^{(m)}$. The representation associated with M_m is defined as:

$$\mathbf{x}_i^m = \mathcal{F}_m\left(\mathbf{r}_i[\Omega_m^r] \| \mathbf{s}_i^{(m)}[\Omega_m^s] \| \mathbf{p}_i^m[\Omega_m^{p\prime}], \boldsymbol{\epsilon}_i^m\right), \tag{3}$$

where $\mathbf{x}_i^m$ denotes the realization of modality node M_m, and $\mathcal{F}_m(\cdot)$ denotes the SCM-based structural mechanism governing the transformation from the selected root, shared, and private representations to $\mathbf{x}_i^m$. The index sets Ω_m^r, Ω_m^s, and $\Omega_m^{p\prime}$ specify the selected dimensions from $\mathbf{r}_i$, $\mathbf{s}_i^{(m)}$, and $\mathbf{p}_i^m$, respectively, while $\boldsymbol{\epsilon}_i^m$ denotes the noise term. By varying the graph configurations and structural mechanisms across SMCMs, we construct synthetic multimodal datasets with diverse correlation structures.

3.3 Fusion Anything Model

FAM is designed to learn transferable correlation patterns through in-context learning over large-scale synthetic multimodal data. Figure 3 presents an overview of the training and inference of our model. During training, each dataset generated with SMCM is partitioned into a labeled context set $\mathcal{D}_{\text{con}}$ and a query set $\mathcal{D}_{\text{q}}$, and FAM is optimized to predict query labels conditioned on the context examples. Training across diverse datasets encourages FAM to encode correlation patterns that generalize beyond individual modality combinations and prediction objectives. During inference, FAM employs labeled context examples from real-world tasks to activate the appropriate multimodal correlation patterns learned during training and predict query labels.

Modality Projector. The modality projector maps heterogeneous modality representations into a common d-dimensional token space for joint multimodal modeling. For sample i, the representation $\mathbf{x}_i^m$ associated with modality M_m is projected into a token sequence:

$$\mathcal{T}_i^m = \mathcal{P}(\mathbf{x}_i^m), \tag{4}$$

where $\mathcal{P}(\cdot)$ denotes the modality projector and $\mathcal{T}_i^m = (\mathbf{t}_{i,\ell}^m)_{\ell=1}^{L_m}$ is a sequence of L_m tokens with $\mathbf{t}_{i,\ell}^m \in \mathbb{R}^d$. Specifically, parallel nonlinear projection heads incorporate gated linear units (GLUs) to selectively regulate feature activations and generate candidate token representations from $\mathbf{x}_i^m$. A cross-attention module uses L_m learnable query embeddings to aggregate the candidate representations into a sequence of L_m modality tokens. The resulting token sequences are concatenated along the sequence dimension across all available modalities, yielding a unified multimodal representation whose length is determined by the available modalities and their respective token counts.

Attention Mechanism. FAM models intra-modality, cross-modality, and sample-level correlations through attention mechanisms over the projected multimodal tokens. Intra-modality attention operates among tokens belonging to the same modality within each sample, capturing relationships among different components of its representation. Cross-modality attention enables interactions among tokens from different modalities within the same sample, integrating consistent and complementary information for prediction. Sample-level attention operates across samples, allowing

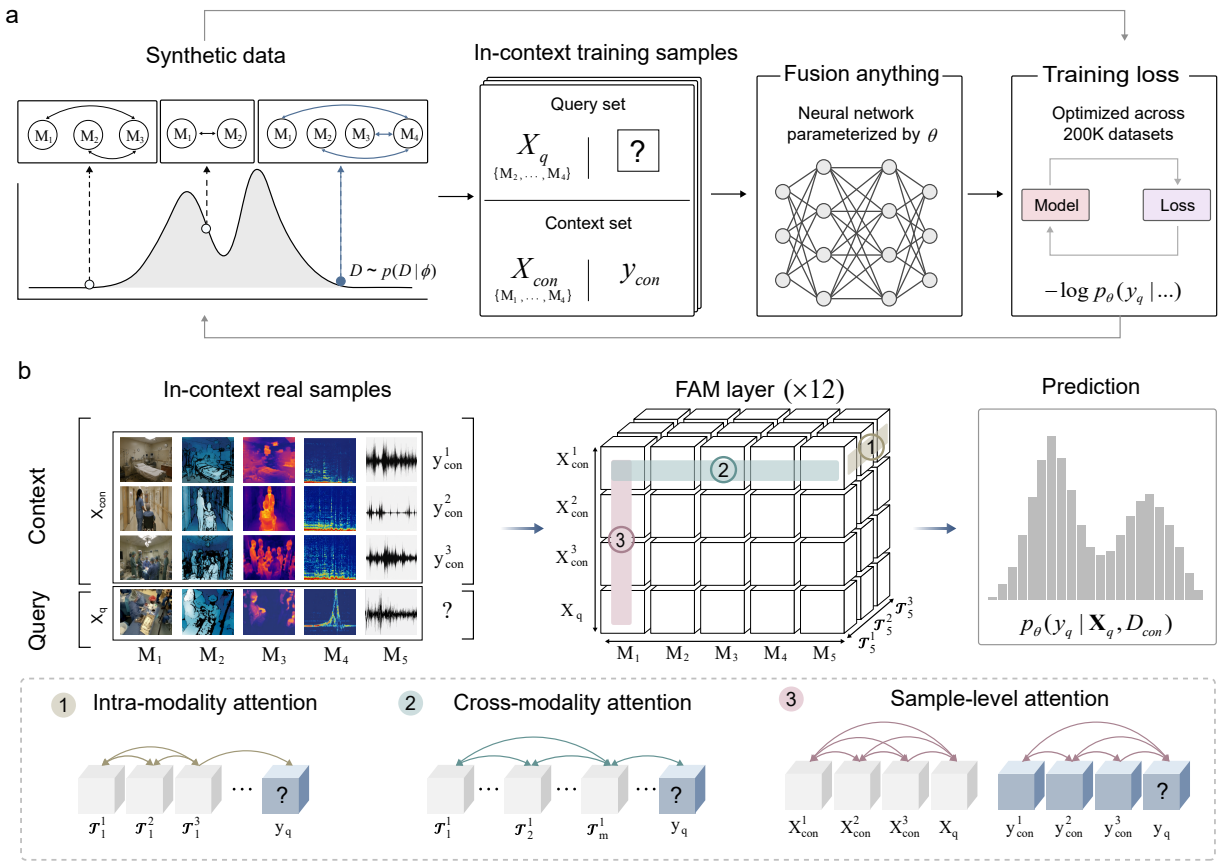


Figure 3: Overview of FAM. (a) Training: synthetic datasets generated by SMCMs are organized to in-context prediction tasks, with query-label guiding the optimization of the proposed FAM across tasks. (b) Inference: given labeled context examples from a real-world multimodal task, FAM predicts labels for unlabeled query samples by leveraging intra-modality, cross-modality, and sample-level attentions.

query tokens to incorporate information from labeled context examples and supporting the inference of predictive relationships relevant to the current task. Together, these attention operations integrate multimodal information within samples and contextual information across samples, enabling context-conditioned multimodal prediction.

Multimodal In-context Learning Objective. The proposed FAM is trained across diverse synthetic multimodal datasets with different causal structures. For each task, the labeled context set $\mathcal{D}_{\text{con}}$ provides task-specific information for identifying relevant multimodal correlations, while the query set $\mathcal{D}_{\text{q}}$ provides the prediction targets used for optimization. The training objective is defined as follows:

$$\mathcal{L}(\theta) = -\mathbb{E}_{\mathcal{D}\sim p(\mathcal{D})}\left[\frac{1}{|\mathcal{D}_{\text{q}}|}\sum_{(\mathbf{X}_i,y_i)\in\mathcal{D}_{\text{q}}}\log p_\theta(y_i \mid \mathbf{X}_i, \mathcal{D}_{\text{con}})\right], \tag{5}$$

where θ denotes the trainable model parameters. By optimizing this objective across synthetic multimodal tasks, FAM learns to exploit contextual examples for prediction under diverse causal structures. We build on the pretrained TabPFN backbone Hollmann et al. [2025] to leverage its existing capacity for modeling feature dependencies and performing context-conditioned prediction. During training, the backbone is updated through Low-Rank Adaptation (LoRA) Hu et al. [2021], together with the modality projector and prediction head, while the remaining pretrained parameters are frozen.

3.4 Inference

After training on synthetic multimodal data, the proposed FAM can be directly applied to real-world datasets. As shown in Figure 3(b), for a real-world dataset $\mathcal{D}$ with M modalities, observations from each modality are first encoded by

pre-trained modality encoders to obtain $\mathbf{x}_i^m$ and subsequently mapped into the unified token space. Since our model concentrates on the modality representations rather than encoder-specific architectures, it remains flexible with respect to the choice of pre-trained encoders. Given a labeled real-world context set $\mathcal{D}_{\text{con}}$ and an unlabeled query input $\mathbf{X}_q$, the model predicts its target according to $p_\theta(y_q \mid \mathbf{X}_q, \mathcal{D}_{\text{con}})$ while keeping all model parameters fixed. This enables direct prediction on novel multimodal datasets with diverse modality combinations and prediction tasks without task-specific optimization.

4 Experiments

4.1 Experimental Settings

Datasets and Evaluation Metrics. We evaluate FAM on a diverse collection of real-world multimodal datasets covering various application domains, such as social media analysis Suryawanshi et al. [2020], Xue et al. [2026], Niu et al. [2016], affective computing Castro et al. [2019], Xu et al. [2022], Livingstone and Russo [2018], Sharma et al. [2020], Schmidt et al. [2018], visual understanding Silberman et al. [2012], Leiva et al. [2020], Song et al. [2015], medical Wu et al. [2023], Weinstein et al. [2013], Rotemberg et al. [2021], Pacheco et al. [2020], Bennett et al. [2018], and remote sensing Xue et al. [2026], Irvin et al. [2020]. The main paper presents evaluation results on 18 datasets, while the Appendix reports the complete results across all 23 datasets, together with detailed descriptions of their modality configurations and data splits. Performance is assessed using Accuracy (ACC) and the Area Under the Receiver Operating Characteristic Curve (AUC).

Comparison Methods. We compare FAM with seven representative supervised multimodal fusion methods, categorized into early, intermediate, and late fusion according to where multimodal information is integrated. Early fusion combines modality features before joint predictive modeling, including Concat Xue et al. [2026] and Early Fusion Transformer (EFT) Liang et al. [2021]. Intermediate fusion combines representations from different modalities within the encoding process, including Multi-to-One (MTO) Li et al. [2021], One-to-Multi (OTM) Lin et al. [2020], and Cross-Attention Fusion (CAF) Lu et al. [2019]. Late fusion combines outputs from individual modality predictors, represented by Logit Summation (LS) Xue et al. [2026] and TMC Han et al. [2022b], which use direct logit summation and uncertainty-aware evidential fusion, respectively.

Model Training. We train our model following the in-context learning paradigm, where each synthetic multimodal dataset is treated as an independent prediction task. During training, we perform approximately $12,000$ optimization steps, with each step sampling 16 synthetic multimodal datasets, resulting in around $192,000$ unique synthetic tasks. The number of samples in each dataset is randomly generated within the range of 512 to $2,048$, and each task is further divided into a context set and a query set, where the query size is fixed to 128 while the remaining samples serve as contextual examples. We optimize the model using the Adam optimizer Kingma and Ba [2015] with a learning rate of 2×10^{-4}, employing a linear warm-up phase followed by cosine annealing for learning-rate scheduling Loshchilov and Hutter [2017].

4.2 Main Results

Figure 4 summarizes the accuracy rankings of the proposed FAM and representative specialized fusion methods across real-world multimodal datasets. No specialized fusion method consistently outperforms the others across all evaluated modality combinations and prediction tasks, with relative rankings varying substantially across datasets. In contrast, FAM achieves strong performance on nearly all evaluated datasets across diverse application domains. The tied or lower rankings of FAM on SIIM-ISIC and Trento reflect closely matched accuracy scores across methods, as shown in the detailed results in the Appendix B.1.

We further present detailed comparisons across the 18 datasets reported in Figure 5. FAM improves ACC by an average of **6.92%** over the mean performance of the specialized fusion methods. On class-imbalanced datasets, such as ForestNet and NYUDv2, we observe improvements in both overall accuracy and minority-class prediction, suggesting that the gains are not confined to majority classes. In addition, specialized fusion methods often exhibit substantial standard deviations on small- to moderate-sized datasets, while FAM consistently achieves competitive or better accuracy. For example, FAM outperforms all specialized methods on GAMMA and MultiOFF, further demonstrating its effectiveness when labeled samples are limited. Detailed results are provided in Appendix B.1.

4.3 Ablation Study

Effectiveness of Multimodal Extension. To evaluate the effectiveness of the proposed FAM in extending context-based prediction toward generalized multimodal scenarios, we compare it with TabPFN under the same input settings. Here,

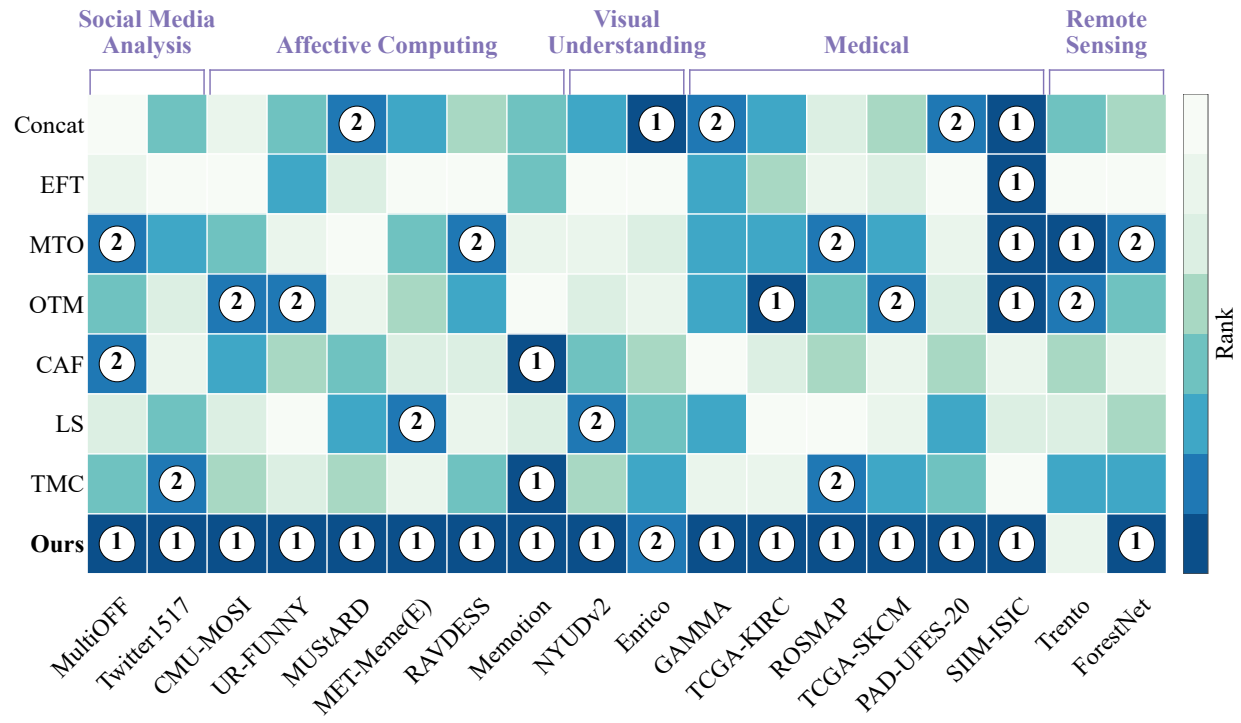


Figure 4: Comparison of accuracy ranking between FAM and representative specialized fusion methods across diverse multimodal datasets. Our model achieves performance comparable to these specialized methods without parameter adaptation.

we first apply PCA to the raw modality representations and project them into the same token space, ensuring that both models receive the same number of input tokens for a fair comparison. As shown in Table 1, our model exhibits improved overall performance across diverse datasets, indicating its enhanced capability in modeling multimodal correlations. These results demonstrate that training over diverse synthetic datasets enables our Fusion Anything model to acquire richer multimodal correlation patterns and effectively extend the prediction capability of the base model to heterogeneous multimodal scenarios. Additional comparison results are provided in the Appendix B.2.

Table 1: Comparison of the original TabPFN and the proposed FAM.

Method	GAMMA		RAVDESS		NYUDv2		MET-Meme(E)		TCGA-SKCM		MultiOFF		TCGA-KIRC		ROSMAP	
	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑
TabPFN	65.00	86.33	57.92	89.21	57.80	90.12	33.65	67.04	55.10	56.32	61.74	62.81	72.73	**79.34**	73.58	**86.35**
Ours	**85.00**	**94.67**	**70.00**	**94.36**	**62.84**	**91.22**	**37.91**	**68.54**	**59.18**	**60.53**	**65.77**	**68.44**	**75.76**	72.31	**76.42**	84.56

Effect of Contextual Examples. To investigate the impact of contextual examples on prediction performance, we evaluate our Fusion Anything model under different context fractions while keeping the query set fixed. Specifically, contextual examples are randomly sampled from the available context pool with three random seeds $\{1, 2, 3\}$ at fractions ranging from 10% to 100%. As shown in Figure 6, our model exhibits an overall positive scaling trend across diverse multimodal datasets. The prediction performance generally improves as more contextual examples become available, while the marginal gains tend to diminish at larger context fractions. This trend indicates that Fusion Anything can effectively leverage contextual examples to identify task-relevant multimodal correlation patterns, while maintaining effective prediction even with limited contextual information. These results further demonstrate the stability of the model's in-context prediction capability across varying context sizes.

Flexibility in the Choice of Pre-trained Encoders. To demonstrate that the proposed FAM is not restricted to particular modality encoders, we evaluate its performance using different pre-trained encoder configurations. As reported in Table 2, our model maintains strong prediction performance, achieving an average improvement of 8.55% over the mean performance of baseline methods across all evaluated settings. This advantage is consistently observed across



Figure 5: Comparison between FAM and specialized fusion models across diverse multimodal datasets.

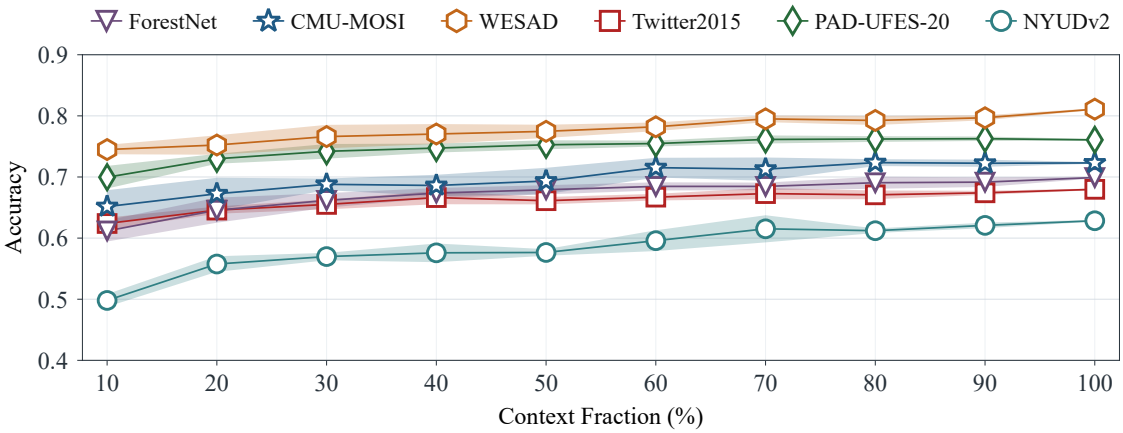


Figure 6: Performance scaling with different fractions of contextual examples.

multiple datasets, suggesting that the learned multimodal correlations can be effectively utilized despite variations in the underlying feature representations. Although the quality of extracted representations inevitably affects the final performance, our model is able to activate and utilize appropriate correlation patterns learned during training, achieving comparable or superior results to baseline methods in most cases. These findings suggest that FAM learns transferable multimodal correlation patterns from synthetic datasets, enabling it to accommodate diverse modality representations without relying on a particular choice of encoders.

Table 2: Comparison under different pre-trained encoder configurations. For each configuration, the best and second-best results among all methods are highlighted in bold and underlined, respectively.

Dataset	Modalities	Combination	Concat	EFT	MTO	CAF	TMC	Avg.	Ours
MultiOFF	Image+Text	DINOv3+Qwen3	58.39	59.96	61.30	61.30	61.07	60.40	**65.77**
		ResNet18+Qwen3	58.61	57.27	59.96	59.06	57.49	58.48	**63.09**
		DINOv3+BERT	62.42	60.40	56.82	59.73	65.10	60.89	**66.44**
RAVDESS	Audio+Visual	WavLM+DINOv3	56.74	26.60	60.83	56.39	56.87	51.49	**70.00**
		Whisper+DINOv3	77.71	32.08	68.82	66.94	**77.99**	64.71	76.04
		WavLM+ResNet18	37.92	18.68	47.08	37.64	35.90	35.44	**58.54**
Twitter2015	Image+Text	ResNet18+BERT	64.77	63.20	66.47	67.12	64.51	65.21	**67.98**
		DINOv3+BERT	64.93	62.84	65.57	64.87	64.13	64.47	**67.31**
		ResNet18+Qwen3	60.88	58.66	64.90	**67.05**	63.36	62.97	65.86

5 Conclusion

In this paper, we propose FAM, a generalized multimodal foundation model toward unified prediction beyond predefined modality combinations and prediction tasks. By constructing large-scale synthetic datasets with diverse causal structures, FAM learns transferable multimodal correlation patterns rather than task-specific fusion strategies. Through multimodal in-context learning, the model leverages contextual examples from unseen datasets to activate task-relevant correlation patterns and perform prediction without additional parameter updates. Extensive experiments demonstrate that our model achieves competitive or superior performance to state-of-the-art specialized fusion models across diverse multimodal benchmarks, while requiring no task-specific optimization at inference time. The proposed FAM provides a promising direction toward supporting future modalities beyond those explored by existing multimodal learning paradigms. Moreover, there are great potential in exploring applications (e.g., embodied intelligence, scientific research) in the future.

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A Dataset Details

GAMMA

Data Source and Content GAMMA Wu et al. [2023] is from the Glaucoma grAding from Multi-Modality imAges challenge. It contains color fundus images and OCT volumes for three-class glaucoma grading.

Link `https://zenodo.org/records/15119049`

Feature Extraction Both the fundus images and OCT volumes are encoded using an ImageNet-pretrained ResNet-18.

Data Splits A stratified split with seed 42 first assigns 80 samples to the non-test partition and 20 samples to the test set. The non-test partition is then divided using another stratified split with seed 10, yielding 64 training, 16 validation, and 20 test samples.

TCGA-KIRC

Data Source and Content TCGA-KIRC Weinstein et al. [2013] contains aligned molecular measurements for kidney renal clear cell carcinoma patients. Six omics views are used for binary survival-status classification.

Link `https://portal.gdc.cancer.gov/projects/TCGA-KIRC`

Feature Extraction CNV, DNA methylation, miRNA, RNA-seq, RPPA, and cell-composition measurements are used directly. Each omics modality is represented using an identity mapping encoder.

Data Splits A fixed non-stratified split with seed 10 is used, containing 92 training, 41 validation, and 33 test patients.

NYUDv2

Data Source and Content The NYU-Depth Dataset V2 (NYUDv2) Silberman et al. [2012] provides RGB and depth images of various indoor scenes captured using a Microsoft Kinect. We use 10 scene categories for classification.

Link `https://cs.nyu.edu/~fergus/datasets/nyu_depth_v2.html`

Feature Extraction An ImageNet-pretrained ResNet-18 is used for the RGB images and another for the depth images.

Data Splits The dataset is split into 381 training, 414 validation, and 654 test samples.

CMU-MOSI

Data Source and Content CMU-MOSI Zadeh et al. [2016] contains opinion video segments with aligned visual, acoustic, and language signals and sentiment-intensity annotations. We use it for binary positive versus non-positive sentiment classification.

Link `https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK`

Feature Extraction We use word-aligned FACET 4.2 visual features, COVAREP acoustic features, and GloVe 840B language features. After removing left padding using the language mask, each modality is averaged over its valid temporal positions.

Data Splits The official split is used, containing 1,283 training, 214 validation, and 686 test segments.

Enrico

Data Source and Content Enrico Leiva et al. [2020] contains mobile UI screenshots and corresponding semantic wireframes for interface-topic classification. The original 20 design topics are merged into ten functional classes.

Link `https://github.com/luileito/enrico`

Feature Extraction The screenshots and semantic wireframes are independently encoded using an ImageNet-pretrained ResNet-18.

Data Splits A non-stratified split with seed 0 first assigns 80% of the samples to the development partition and 20% to the test set. The development partition is then divided using another non-stratified split with seed 0, yielding 992 training, 176 validation, and 292 test interfaces.

ROSMAP

Data Source and Content ROSMAP Bennett et al. [2018]contains multi-omics data from post-mortem brain tissue collected by the Religious Orders Study and Memory and Aging Project. We use it for binary classification of Alzheimer's disease and normal controls.

Link `https://github.com/txWang/MOGONET`

Feature Extraction Each tabular omics modality is encoded using an independent identity mapping encoder.

Data Splits The released split first assigns 245 samples to the non-test partition and 106 samples to the test set. The non-test partition is then divided using a stratified split with seed 10, yielding 196 training, 49 validation, and 106 test samples.

UR-FUNNY

Data Source and Content UR-FUNNY Hasan et al. [2019] contains aligned language, acoustic, and visual observations from public speeches for binary multimodal humor detection.

Link `https://github.com/ROC-HCI/UR-FUNNY`

Feature Extraction We use the released GloVe language embeddings, COVAREP acoustic features, and OpenFace visual features. The specialized models use temporal means, while the PCA48 TabPFN pipeline concatenates the temporal mean and standard deviation of each modality.

Data Splits The official split is used, containing 7,614 training, 980 validation, and 994 test samples.

TCGA-SKCM

Data Source and Content TCGA-SKCM Weinstein et al. [2013] contains aligned molecular measurements for skin cutaneous melanoma patients. Six omics views from complete-case patients are used for binary survival-status classification.

Link `https://portal.gdc.cancer.gov/projects/TCGA-SKCM`

Feature Extraction CNV, DNA methylation, miRNA, RNA-seq, RPPA, and cell-composition measurements are used directly. Each omics modality is represented using an identity mapping encoder.

Data Splits A fixed non-stratified split with seed 10 is used, containing 137 training, 61 validation, and 49 test patients.

MultiOFF

Data Source and Content MultiOFF Suryawanshi et al. [2020] contains image–text memes for binary offensive-content detection.

Link `https://github.com/bharathichezhiyan/Multimodal-Meme-Classification-Identifying-Offensive-Content-i`

Feature Extraction Images are encoded using a frozen DINOv3 ViT-B/16, and text is encoded using Qwen3-Embedding-0.6B.

Data Splits The official split is used, containing 445 training, 149 validation, and 149 test posts.

MET-Meme(E)

Data Source and Content The English subset of MET-Meme Xu et al. [2022] contains image–text memes for seven-class sentiment classification.

Link `https://github.com/liaolianfoka/MET-Meme-A-Multi-modal-Meme-Dataset-Rich-in-Metaphors`

Feature Extraction Meme images are encoded using an ImageNet-pretrained ResNet-18, and text is encoded using BERT-base-uncased with attention-mask mean pooling.

Data Splits A non-stratified 7:1:2 random split with seed 42 is used, containing 737 training, 105 validation, and 211 test samples.

RAVDESS

Data Source and Content The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) Livingstone and Russo [2018] contains recordings from 24 professional actors. We use its speech subset and pair each official audio-only recording with the corresponding video-only recording for eight-class emotion recognition: neutral, calm, happy, sad, angry, fearful, disgust, and surprised.

Link `https://zenodo.org/records/1188976`

Feature Extraction Audio recordings are encoded using WavLM Base+, while eight uniformly sampled video frames are encoded using DINOv3 ViT-B/16 and averaged to obtain visual representations.

Data Splits We use a fixed actor-disjoint split. Actors 4, 10, 16, and 22 form the 240 validation utterances; actors 5, 6, 11, 12, 17, 18, 23, and 24 form the 480 test utterances; and the remaining actors form the 720 training utterances.

Memotion

Data Source and Content Memotion Sharma et al. [2020] contains memes with transcribed text and affective annotations. We use its binary humor task, treating not-funny as the negative class and merging funny, very funny, and hilarious into the positive class.

Link `https://www.kaggle.com/datasets/williamscott701/memotion-dataset-7k`

Feature Extraction Meme images are encoded using DINOv3 ViT-B/16, while transcribed text is encoded using Qwen3-Embedding-0.6B.

Data Splits The processed dataset contains 6,831 samples. A fixed non-stratified 8:1:1 split with seed 42 is used, containing 5,465 training, 683 validation, and 683 test samples.

Twitter1517

Data Source and Content Twitter1517 Xue et al. [2026] contains paired tweet text and images for binary image–text correlation classification.

Link `https://github.com/code-chendl/HFIR`

Feature Extraction Images are encoded using a frozen DINOv3 ViT-B/16, and text is encoded using Qwen3-Embedding-0.6B.

Data Splits A stratified 7:1:2 split with seed 42 is used, containing 3,270 training, 467 validation, and 935 test samples.

Trento

Data Source and Content Trento Xue et al. [2026] covers a rural area south of Trento, Italy. It contains co-registered hyperspectral imagery and a LiDAR-derived digital surface model for six-class land-cover classification.

Link `https://github.com/tyust-dayu/Trento/tree/b4afc449ce5d6936ddc04fe267d86f9f35536afd`

Feature Extraction A frozen HyperFree-b model encodes 5×5 hyperspectral patches, while a frozen DOFA ViT-Base model encodes the corresponding 5×5 digital-surface-model patches.

Data Splits A stratified 7:1:2 split with seed 42 first defines 24,170 non-test and 6,044 test samples. The non-test partition is then divided using a stratified 8:2 split with seed 10, yielding 19,336 training, 4,834 validation, and 6,044 test samples.

ForestNet

Data Source and Content ForestNet Irvin et al. [2020] contains satellite observations of forest-loss events in Indonesia together with expert annotations of their direct drivers. We use the four merged classes: plantation, smallholder agriculture, grassland/shrubland, and other.

Link `https://stanfordmlgroup.github.io/projects/forestnet/`

Feature Extraction RGB imagery and three-channel SRTM data are encoded using a shared frozen ImageNet-pretrained ResNet-152. Latitude, longitude, and year are retained as a three-dimensional structured modality using identity mapping.

Data Splits The processed official split is retained without additional randomization, containing 1,616 training, 473 validation, and 668 test samples.

MUStARD

Data Source and Content MUStARD Castro et al. [2019]contains 690 aligned text, audio, and video utterances from television shows for binary sarcasm detection.

Link `https://github.com/soujanyaporia/MUStARD`

Feature Extraction Text is represented by the mean of the last four BERT-base-uncased CLS layers. The officially released 283-dimensional librosa audio features are averaged over temporal bins, and ImageNet-pretrained ResNet-152 pool5 video features are averaged over frames.

Data Splits Fold 1 of the official speaker-dependent stratified five-fold split is used. Its 552 non-test utterances are divided using a stratified 8:2 split with seed 10, yielding 441 training, 111 validation, and 138 test utterances.

PAD-UFES-20

Data Source and Content PAD-UFES-20 Pacheco et al. [2020] contains 2,298 smartphone clinical images from 1,373 patients, together with patient metadata for six-class skin-lesion diagnosis.

Link `https://data.mendeley.com/datasets/zr7vgbcyr2`

Feature Extraction Clinical images are encoded using a frozen ImageNet-1K V2 ResNet-50. Patient metadata is retained as a structured modality after missing-value handling and categorical encoding fitted on each outer non-test partition.

Data Splits A patient-grouped stratified five-fold split with seed 42 is used. Within each outer non-test partition, a patient-grouped stratified five-fold split with seed 10 defines the validation set, yielding 1,470–1,471 training, 368 validation, and 459–460 test samples per fold. Results are averaged over the five outer folds.

SIIM-ISIC

Data Source and Content SIIM-ISIC Rotemberg et al. [2021] is from the 2020 SIIM-ISIC Melanoma Classification challenge. It contains dermoscopic images of skin lesions and patient-level metadata for binary melanoma classification.

Link `https://www.kaggle.com/competitions/siim-isic-melanoma-classification/data`

Feature Extraction Dermoscopic images are encoded using an ImageNet-pretrained ResNet-18. Sex, approximate age, and anatomical site are retained as a structured modality using identity mapping after missing-value handling and categorical encoding fitted on the non-test partition.

Data Splits A fixed non-stratified 8:1:1 split with seed 42 is used, containing 26,502 training, 3,312 validation, and 3,312 test samples.

MELD

Data Source and Content The Multimodal EmotionLines Dataset (MELD) Poria et al. [2019] contains 13,708 utterances from multi-party conversations in the television series Friends, with aligned text, audio, and video. We use the three-class sentiment-classification task.

Link `https://github.com/declare-lab/MELD`

Feature Extraction We use the MultiBench-released utterance-level GloVe-average text features and feature-selected acoustic features directly through identity mapping encoders.

Data Splits The MultiBench split is used, containing 9,989 training, 1,109 validation, and 2,610 test utterances.

WESAD

Data Source and Content WESAD Schmidt et al. [2018] is a wearable stress-and-affect dataset recorded from 15 subjects using chest- and wrist-worn sensors. We retain the baseline, stress, and amusement states for three-class classification.

Link `https://archive.ics.uci.edu/dataset/465/wesad`

Feature Extraction Chest channels are jointly encoded using a frozen MOMENT-1-small, while the four wrist signals are encoded separately and averaged. Embeddings from consecutive 512-sample blocks are averaged within each 60-second window.

Data Splits Each contiguous retained-state interval is divided into non-overlapping 60-second windows, yielding 535 windows. A deterministic 15-fold leave-one-subject-out split is used. In each fold, one subject forms the test set, another forms the validation set according to a balanced schedule with seed 10, and the remaining 13 form the training set, yielding 462–465 training, 35–37 validation, and 35–37 test windows. Results are averaged over the 15 folds.

Twitter2015

Data Source and Content Twitter2015 Xue et al. [2026] originates from SemEval tasks and pairs tweets with relevant images for three-class multimodal aspect-based sentiment analysis.

Link `https://archive.org/details/twitterstream`

Feature Extraction Images are encoded using a frozen ImageNet-pretrained ResNet-18. The two text fields are concatenated and encoded using BERT-base-uncased with attention-mask mean pooling.

Data Splits The official split is used, containing 3,179 training, 1,122 validation, and 1,037 test samples.

SUN RGB-D

Data Source and Content SUN RGB-D Song et al. [2015] is a large-scale indoor-scene benchmark containing paired RGB and depth images captured using multiple RGB-D sensors. We use the fixed 19-category subset and merge the original categories into eight semantic classes: bathroom, residential, education, workspace, meeting, dining, service, and furniture.

Link `https://rgbd.cs.princeton.edu/`

Feature Extraction The RGB images and depth maps are independently encoded using an ImageNet-pretrained ResNet-18.

Data Splits The fixed MultiBench directory split first assigns 4,845 samples to the non-test partition and 4,659 samples to the test set. The non-test partition is then divided using a stratified 8:2 split with seed 10, yielding 3,876 training, 969 validation, and 4,659 test samples.

MVSA-Single

Data Source and Content MVSA-Single Niu et al. [2016] contains image–text posts collected from Twitter for three-class sentiment classification: positive, neutral, and negative. The Single variant contains posts with consistent sentiment labels across annotators.

Link `https://www.kaggle.com/datasets/vincemarcs/mvsasingle`

Feature Extraction Images are encoded using a frozen ImageNet-pretrained ResNet-18, and text is encoded using BERT-base-uncased with attention-mask mean pooling.

Data Splits The provided JSONL split is used, containing 1,555 training, 518 validation, and 519 test posts.

B More Experimental Details

B.1 Supplementary Results for the Main Experiment

Table 3 provides the complete quantitative results corresponding to the main experiment. For all specialized fusion methods, the reported results are averaged over three independent training runs with different random seeds $(1, 2, 3)$, as these methods require task-specific parameter optimization. In contrast, FAM performs prediction through parameter-free multimodal in-context inference without adaptation. Therefore, its results are deterministic under the same evaluation setting and are reported directly without variance estimation.

Table 3: Comparison results across different multimodal datasets. Results of the specialized fusion methods are reported as mean ± standard deviation, with the standard deviation shown below the mean in the same secondary-line font size as the ▲/▼ marker. The marker below each Ours ACC or AUC reports its increase/decrease relative to the mean of the corresponding metric over the specialized fusion methods.

Dataset	Concat		EFT		MTO		OTM		CAF		LS		TMC		**Ours**	
	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑	ACC↑	AUC↑
GAMMA	65.00 (± 5.00)	85.59 (± 1.58)	63.33 (± 16.07)	74.22 (± 15.70)	63.33 (± 12.58)	83.11 (± 8.20)	63.33 (± 10.41)	76.44 (± 15.28)	58.33 (± 5.77)	77.48 (± 8.23)	63.33 (± 7.64)	77.07 (± 17.49)	60.00 (± 5.00)	83.56 (± 3.11)	**85.00** (▲22.62)	**94.67** (▲15.03)
TCGA-KIRC	71.72 (± 4.63)	70.11 (± 8.51)	70.71 (± 4.63)	72.59 (± 9.62)	71.72 (± 4.63)	66.94 (± 5.74)	**75.76** (± 9.09)	65.98 (± 4.22)	69.70 (± 6.06)	**73.55** (± 5.28)	61.62 (± 9.74)	62.53 (± 5.17)	63.64 (± 3.03)	67.91 (± 3.56)	**75.76** (▲6.49)	72.31 (▲3.80)
NYUDv2	55.45 (± 0.64)	84.03 (± 0.37)	30.73 (± 2.31)	62.85 (± 3.27)	42.86 (± 1.79)	67.11 (± 7.45)	52.34 (± 3.24)	79.67 (± 4.07)	54.18 (± 1.39)	83.93 (± 1.15)	55.81 (± 1.86)	83.39 (± 1.65)	53.57 (± 0.64)	83.42 (± 0.97)	**62.84** (▲13.56)	**91.22** (▲13.44)
CMU-MOSI	57.43 (± 1.77)	75.55 (± 0.59)	53.60 (± 2.62)	62.00 (± 11.40)	67.06 (± 1.27)	72.71 (± 2.45)	69.83 (± 1.77)	72.89 (± 2.87)	67.69 (± 1.53)	75.10 (± 1.08)	61.61 (± 3.08)	71.92 (± 4.93)	65.45 (± 1.91)	72.42 (± 4.90)	**72.30** (▲9.06)	**79.58** (▲7.78)
Enrico	**52.97** (± 0.71)	**85.65** (± 0.57)	26.71 (± 2.99)	63.37 (± 1.57)	47.15 (± 4.85)	76.15 (± 5.96)	45.66 (± 1.20)	78.63 (± 2.82)	49.77 (± 3.33)	83.64 (± 1.46)	50.91 (± 4.23)	85.38 (± 2.32)	51.94 (± 1.62)	85.31 (± 1.15)	52.74 (▲6.30)	85.06 (▲5.32)
ROSMAP	66.67 (± 3.93)	80.81 (± 4.27)	63.52 (± 10.60)	78.37 (± 7.63)	72.96 (± 3.57)	78.85 (± 2.66)	70.44 (± 4.75)	81.47 (± 1.92)	67.92 (± 1.63)	83.45 (± 0.36)	61.64 (± 9.68)	65.54 (± 11.42)	72.96 (± 3.31)	81.97 (± 4.77)	**76.42** (▲8.40)	**84.56** (▲5.93)
UR-FUNNY	60.76 (± 0.27)	66.29 (± 0.25)	60.83 (± 0.52)	66.20 (± 0.22)	60.63 (± 0.15)	64.79 (± 1.68)	61.67 (± 1.32)	**66.67** (± 1.39)	60.73 (± 0.25)	66.06 (± 0.22)	59.29 (± 1.75)	65.23 (± 1.50)	60.66 (± 0.46)	65.56 (± 0.13)	**61.97** (▲1.32)	65.55 (▼0.28)
TCGA-SKCM	54.42 (± 1.18)	52.22 (± 2.19)	53.74 (± 2.36)	52.11 (± 1.86)	56.46 (± 5.14)	53.39 (± 6.30)	58.50 (± 2.36)	52.69 (± 2.33)	52.38 (± 10.47)	51.99 (± 8.87)	52.38 (± 8.50)	50.41 (± 5.04)	56.46 (± 7.73)	54.68 (± 14.27)	**59.18** (▲4.27)	**60.53** (▲8.03)
MultiOFF	58.39 (± 1.34)	61.82 (± 2.19)	59.96 (± 2.71)	60.45 (± 3.46)	61.30 (± 3.44)	60.12 (± 4.66)	61.07 (± 2.93)	62.73 (± 4.23)	61.30 (± 0.39)	61.39 (± 2.70)	60.40 (± 2.42)	61.45 (± 5.04)	61.07 (± 4.84)	61.05 (± 3.28)	**65.77** (▲5.27)	**68.44** (▲7.15)
MET-Meme(E)	33.81 (± 4.30)	**69.01** (± 1.34)	19.91 (± 4.10)	54.82 (± 3.45)	32.70 (± 2.51)	58.78 (± 0.12)	32.39 (± 0.27)	62.63 (± 1.90)	32.23 (± 5.77)	61.07 (± 3.92)	35.07 (± 2.46)	66.13 (± 4.86)	29.86 (± 2.64)	63.39 (± 5.70)	**37.91** (▲7.06)	68.54 (▲6.28)
RAVDESS	56.74 (± 1.88)	90.03 (± 0.59)	26.60 (± 1.68)	68.77 (± 3.31)	60.83 (± 5.83)	87.30 (± 3.56)	59.24 (± 4.76)	87.79 (± 0.87)	56.39 (± 2.84)	88.61 (± 0.96)	55.63 (± 1.10)	89.90 (± 0.52)	56.87 (± 1.67)	89.77 (± 0.27)	**70.00** (▲16.81)	**94.36** (▲8.34)
Memotion	78.14 (± 0.08)	**59.26** (± 0.82)	78.14 (± 0.08)	54.69 (± 1.88)	77.35 (± 0.66)	56.74 (± 2.12)	76.87 (± 1.54)	55.42 (± 3.42)	**78.18** (± 0.00)	58.13 (± 0.46)	78.04 (± 0.15)	57.52 (± 1.60)	**78.18** (± 0.15)	59.06 (± 0.18)	**78.18** (▲0.34)	57.25 (▼0.01)
Twitter1517	76.15 (± 0.93)	63.50 (± 2.10)	75.22 (± 0.81)	59.44 (± 0.74)	76.22 (± 0.16)	61.37 (± 1.39)	76.08 (± 0.65)	59.92 (± 2.56)	76.01 (± 1.01)	64.36 (± 0.77)	76.15 (± 0.67)	62.52 (± 2.61)	76.43 (± 0.38)	60.57 (± 4.92)	**76.47** (▲0.43)	**65.87** (▲4.20)
Trento	98.25 (± 0.06)	99.90 (± 0.01)	91.86 (± 3.67)	98.01 (± 1.29)	**98.75** (± 0.11)	99.93 (± 0.06)	98.72 (± 0.03)	**99.95** (± 0.02)	98.17 (± 0.18)	99.94 (± 0.01)	98.02 (± 0.07)	99.87 (± 0.02)	98.33 (± 0.15)	99.93 (± 0.00)	97.75 (▲0.31)	99.89 (▲0.25)
ForestNet	47.36 (± 0.91)	55.44 (± 1.33)	46.76 (± 0.17)	58.32 (± 4.33)	66.52 (± 0.57)	81.21 (± 1.64)	60.53 (± 4.64)	76.78 (± 3.86)	47.31 (± 1.50)	60.32 (± 4.08)	47.36 (± 1.06)	56.35 (± 1.59)	65.37 (± 3.72)	83.15 (± 3.04)	**69.91** (▲15.45)	**87.22** (▲19.85)
MUStARD	71.98 (± 2.21)	79.52 (± 2.93)	69.32 (± 1.11)	75.53 (± 2.49)	65.94 (± 1.92)	74.48 (± 1.66)	66.43 (± 6.33)	72.96 (± 2.39)	71.26 (± 2.54)	78.08 (± 1.49)	71.74 (± 5.93)	78.93 (± 2.95)	70.77 (± 1.67)	**79.80** (± 0.18)	**72.46** (▲2.83)	78.19 (▲1.15)
PAD-UFES-20	74.73 (± 0.24)	91.87 (± 0.07)	66.02 (± 1.70)	77.66 (± 1.63)	71.54 (± 0.66)	87.53 (± 0.80)	71.86 (± 0.16)	87.94 (± 0.69)	72.40 (± 0.89)	90.58 (± 0.50)	74.38 (± 0.61)	91.92 (± 0.23)	74.09 (± 0.36)	90.99 (± 0.14)	**76.11** (▲3.96)	**92.88** (▲4.53)
SIIM-ISIC	**97.83** (± 0.00)	59.67 (± 6.23)	**97.83** (± 0.00)	55.88 (± 3.97)	**97.83** (± 0.00)	57.55 (± 15.52)	**97.83** (± 0.00)	57.34 (± 11.89)	97.78 (± 0.05)	66.31 (± 21.18)	97.81 (± 0.03)	66.55 (± 25.69)	97.75 (± 0.11)	57.68 (± 13.75)	**97.83** (▲0.02)	**84.46** (▲24.32)
MELD	62.20 (± 0.16)	75.93 (± 0.13)	58.10 (± 0.83)	71.02 (± 1.30)	64.20 (± 1.70)	78.77 (± 1.62)	**67.29** (± 0.12)	**80.77** (± 0.64)	65.34 (± 0.27)	78.98 (± 0.68)	61.76 (± 1.03)	75.90 (± 0.19)	63.88 (± 0.74)	77.81 (± 0.12)	66.21 (▲2.96)	80.46 (▲3.43)
WESAD	54.68 (± 1.13)	63.99 (± 1.32)	71.76 (± 3.47)	75.57 (± 2.10)	72.64 (± 0.37)	81.91 (± 1.49)	71.98 (± 1.63)	83.46 (± 1.02)	55.66 (± 2.46)	73.13 (± 10.26)	55.75 (± 2.46)	69.38 (± 5.35)	57.43 (± 4.30)	69.11 (± 5.91)	**81.10** (▲18.26)	**92.12** (▲18.32)
Twitter2015	64.77 (± 0.79)	76.56 (± 1.43)	63.20 (± 1.03)	62.71 (± 6.26)	66.47 (± 0.92)	77.03 (± 3.04)	67.60 (± 0.93)	79.24 (± 0.51)	67.12 (± 0.93)	79.75 (± 1.98)	64.35 (± 0.90)	76.98 (± 1.16)	64.51 (± 1.67)	73.90 (± 4.28)	**67.98** (▲2.55)	**80.13** (▲4.96)
SUN RGB-D	**59.35** (± 0.55)	**90.09** (± 0.13)	47.66 (± 4.67)	80.48 (± 0.95)	45.83 (± 3.90)	80.43 (± 1.22)	58.26 (± 2.18)	88.16 (± 1.33)	55.49 (± 0.53)	88.66 (± 0.26)	58.52 (± 0.34)	89.92 (± 0.07)	58.02 (± 0.25)	89.57 (± 0.07)	57.95 (▲3.22)	89.63 (▲2.87)
MVSA-Single	75.92 (± 2.03)	89.83 (± 0.57)	75.85 (± 1.25)	83.37 (± 2.04)	75.53 (± 0.51)	86.70 (± 1.24)	76.24 (± 1.39)	87.37 (± 1.33)	78.10 (± 1.45)	90.41 (± 1.39)	76.24 (± 1.64)	90.39 (± 0.62)	76.43 (± 1.73)	90.48 (± 0.15)	**78.23** (▲1.90)	**92.02** (▲3.66)

B.2 Supplementary Results for the Effectiveness of Multimodal Extension

Table 4 provides the complete comparison between TabPFN and FAM across diverse real-world datasets. Overall, our model achieves stronger performance across a broad range of multimodal prediction tasks, with improvements observed over heterogeneous modality combinations. These results demonstrate that the proposed model effectively broadens the model's capability to capture and utilize predictive correlations among heterogeneous modalities.

Table 4: Comparison of TabPFN and our model across different multimodal datasets. The ▲/▼ marker indicates the increase/decrease of Fusion Anything relative to the corresponding TabPFN result in percentage points. Ours results that match or exceed TabPFN are shown in bold.

Dataset	TabPFN			**Ours**		
	ACC↑	AUC↑	F1↑	ACC↑	AUC↑	F1↑
GAMMA	65.00	86.33	48.48	**85.00** (▲ 20.00)	**94.67** (▲ 8.33)	**78.02** (▲ 29.54)
TCGA-KIRC	72.73	79.34	61.18	**75.76** (▲ 3.03)	72.31 (▼ 7.02)	**69.44** (▲ 8.26)
NYUDv2	57.80	90.12	50.51	**62.84** (▲ 5.04)	**91.22** (▲ 1.10)	**57.66** (▲ 7.15)
CMU-MOSI	71.28	79.03	70.57	**72.30** (▲ 1.02)	**79.58** (▲ 0.56)	**71.84** (▲ 1.27)
Enrico	51.37	84.49	37.29	**52.74** (▲ 1.37)	**85.06** (▲ 0.56)	**41.25** (▲ 3.96)
ROSMAP	73.58	86.35	73.55	**76.42** (▲ 2.84)	84.56 (▼ 1.78)	**76.36** (▲ 2.81)
UR-FUNNY	63.18	67.74	63.00	61.97 (▼ 1.21)	65.55 (▼ 2.19)	61.78 (▼ 1.22)
TCGA-SKCM	55.10	56.32	54.63	**59.18** (▲ 4.08)	**60.53** (▲ 4.21)	**56.09** (▲ 1.46)
MultiOFF	61.74	62.81	47.30	**65.77** (▲ 4.03)	**68.44** (▲ 5.63)	**58.60** (▲ 11.30)
MET-Meme(E)	33.65	67.04	18.01	**37.91** (▲ 4.26)	**68.54** (▲ 1.51)	**23.87** (▲ 5.86)
RAVDESS	57.92	89.21	57.79	**70.00** (▲ 12.08)	**94.36** (▲ 5.16)	**69.77** (▲ 11.98)
Memotion	78.18	58.20	43.88	**78.18** (▲ +0.00)	57.25 (▼ 0.95)	**43.88** (▲ +0.00)
Twitter1517	76.04	62.24	44.06	**76.47** (▲ 0.43)	**65.87** (▲ 3.63)	**44.21** (▲ 0.15)
Trento	98.58	99.95	97.85	97.75 (▼ 0.83)	99.89 (▼ 0.06)	96.11 (▼ 1.74)
ForestNet	68.86	89.50	64.48	**69.91** (▲ 1.05)	87.22 (▼ 2.28)	**65.94** (▲ 1.46)
MUStARD	73.19	79.81	73.19	72.46 (▼ 0.73)	78.19 (▼ 1.63)	72.46 (▼ 0.73)
PAD-UFES-20	76.50	93.65	62.71	76.11 (▼ 0.39)	92.88 (▼ 0.77)	**64.04** (▲ 1.33)
SIIM-ISIC	97.83	83.69	49.45	**97.83** (▲ +0.00)	**84.46** (▲ 0.77)	**49.45** (▲ +0.00)
MELD	66.82	79.71	61.44	66.21 (▼ 0.61)	**80.46** (▲ 0.75)	60.81 (▼ 0.63)
WESAD	81.62	93.59	74.82	81.10 (▼ 0.52)	92.12 (▼ 1.48)	72.66 (▼ 2.16)
Twitter2015	66.73	79.28	56.14	**67.98** (▲ 1.25)	**80.13** (▲ 0.85)	55.97 (▼ 0.17)
SUN RGB-D	58.70	90.51	55.44	57.95 (▼ 0.75)	89.63 (▼ 0.88)	**57.41** (▲ 1.97)
MVSA-Single	79.58	91.88	76.76	78.23 (▼ 1.35)	**92.02** (▲ 0.14)	74.76 (▼ 2.00)